

Chemistry, 2018 EC, entrance exam, version 1

Grade 12 · Chemistry · 2018 E.C.

Time allowed: 120 minutes**Questions:** 65**Total marks:** 65

Name _____ Grade _____ ID No. _____ Date _____

1. Who used the 'cathode ray tube' to discover the electron and determine its charge to mass ratio? (1)
A. Robert. A. Millikan
B. James Chadwick
C. J.J. Thomson
D. Ernest Rutherford
2. Among the following postulates of Dalton's atomic theory, the one that was modified in the modern atomic theory is (1)
A. Atoms are indivisible
B. Atoms are the smallest particle of all elements
C. Atoms of more than one element combine to form compounds
D. Atoms are indestructible
3. Which of the following statements can be used as an example to describe the law of definite proportion? (1)
A. 48g of magnesium ribbon burns in excess oxygen to form 80g of magnesium oxide
B. The mass in gram of oxygen to phosphorus in P_4O_{10} to that in P_4O_6 is in 3:2 ratio
C. White chalk consists of the elements Calcium, carbon and oxygen which are combined in the mass ratio of 10:3:12 regardless of its source.
D. Water is a colorless liquid at room temperature
4. The statement 'The properties of the elements are periodic functions of their atomic masses' is the (1)
A. Law of mass action
B. Mendeleev's periodic law
C. Modern periodic law
D. Law of definite proportion
5. What was the main conclusion of the Rutherford Gold Foil Experiment conducted by Ernest Rutherford? (1)
A. Electrons carry a negative charge.
B. A proton is a hydrogen atom without an electron.
C. Electrons move around the nucleus in fixed orbits.
D. Most of the mass and all the positive charge of an atom are concentrated in a very small nucleus
6. The number of protons, neutrons and electrons respectively in Cd^{2+} are _____. (1)
A. 48, 64, 48
B. 48, 64, 46
C. 48, 64, 50
D. 48, 63, 46
7. To which group and periods of the periodic table does an element with atomic number 33 belong? (1)
A. Group IIIB and period 4
B. Group IIIA and period 4
C. Group VA and period 4
D. Group IV and period 4

- 8.** An element X has two naturally occurring isotopes. The information about these isotopes is given in the table below. Isotope X-40 (Atomic Mass 40 u, Natural Abundance 80%) and Isotope X-44 (Atomic Mass 44 u, Natural Abundance 20%). What is the average atomic mass of element X? (1)

A. 40.8 u	B. 41.5 u
C. 40.0 u	D. 42.0 u

9. In which of the following blocks of the periodic table does an element with atomic number 16 located? (1)

A. s	B. p
C. d	D. f

10. All of the following compounds contain ionic and covalent bonds except: (1)

A. NH ₄ Cl	B. CaCO ₃
C. NH ₃ BF ₃	D. Na ₃ PO ₄

11. Which of the following molecules considered to be a polar molecule? (1)

A. NH ₃	B. BF ₃
C. CO ₂	D. CCl ₄

12. Which one of the following molecules does not contain dative bond? (1)

A. H ₃ O ⁺	B. NH ₃
C. NH ₃ BF ₃	D. NH ₄ ⁺

13. The attraction force which is created by a temporary redistribution of electrons in neighboring atom is called? (1)

A. Hydrogen bond	B. London dispersion
C. Dipole-dipole	D. Ionic bond

14. Which one of the following does the sulphur atom have the lowest oxidation number? (1)

A. SO ₃	B. S ₂ F ₁₀
C. H ₂ SO ₃	D. H ₂ S

15. The concept like dissolves like is illustrated by which of the following? (1)

A. NaCl is more soluble in CCl ₄ than water	B. I ₂ is more soluble in CCl ₄ than water
C. CuSO ₄ is more soluble in CCl ₄ than water	D. CCl ₄ soluble in water

16. What is the mole fraction of methanol (CH₃OH) in a mixture of equal mass of methanol and water? (1)

A. 0.36	B. 0.64
C. 0.53	D. 0.44

17. Which of the following acid is a monoprotic acid? (1)

A. H ₂ SO ₄	B. CH ₃ COOH
C. H ₃ PO ₄	D. H ₂ CO ₃

18. Student A and Student B are grade 12 students. They carry out laboratory experiment on detection of metallic ions in different salts as per their laboratory assistance intended them to do so. Student A achieved yellow color after he carry out the flame tests of different salts whereas Student B obtained green color from the same tests on different salts. Which ions did Student A and Student B detected respectively? (1)

A. Ca ²⁺ & K ⁺	B. Na ⁺ & Ba ²⁺
C. Ca ²⁺ & Mg ²⁺	D. Ba ²⁺ & Li ⁺

19. Which of the following is an example of double displacement reaction? (1)
- A. $\text{Cl}_2 + 2\text{KI} \rightarrow 2\text{KCl} + \text{I}_2$ B. $\text{NaHCO}_3 \rightarrow \text{Na}_2\text{CO}_3 + \text{CO}_2 + \text{H}_2\text{O}$
C. $\text{Na}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{NaOH}$ D. $\text{NaCl} + \text{AgNO}_3 \rightarrow \text{AgCl} + \text{NaNO}_3$
20. Which of the following is correct statement about endothermic reaction? (1)
- A. Heat energy is written in the right side of the equation
B. Enthalpy change of the reaction has a negative value
C. Chemical reaction that releases heat energy to the surrounding.
D. Heat is absorbed to the system from the surrounding.
21. The two gaseous elements that can be obtained from atmosphere by fractional distillation of liquid air are: (1)
- A. Nitrogen and hydrogen B. Oxygen and hydrogen
C. Nitrogen and oxygen D. Ammonia and carbon dioxide
22. According to Markovnikov's rule, the alcohol produced from propene, $\text{CH}_2=\text{CH}-\text{CH}_3$, during the addition of water in the presence of acid catalyst is (1)
- A. 2-propanol B. 1-propanol
C. Hexanol D. 2-hexanol
23. Which of the following molecules does not exhibit geometrical isomerism? (1)
- A. 2-butene B. 1-butene
C. 3-hexene D. 1,2-dichloro ethene
24. Among the following methods, which one can be used to prepare alkyne in the laboratory? (1)
- A. Dehydration of alcohols with concentrated H_2SO_4
B. Hydrogenation of alkene with a metal catalyst.
C. Dehydrohalogenation of alkyl halides with a base
D. Dehydrohalogenation of Vicinal dihalides with a base.
25. What is the wavelength of a photon emitted when a hydrogen atom undergoes a transition from $n=5$ to $n=2$? (1)
- A. 434 mm B. 434 pm
C. 434 nm D. 434 μm
26. For the four quantum numbers needed to describe an electron, which of these combinations are not possible? (1)
- A. (1, 0, 0, +1/2) B. (2, 1, 1, -1/2)
C. (3, 2, -2, -1/2) D. (1, 0, -1, +1/2)
27. Which of the following is the most stable electronic configuration for the element 24Cr? (1)
- A. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^4$ B. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$
C. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$ D. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4 4s^2$
28. Which pairs of molecules is geometrically similar? (1)
- A. CF_4 and SF_4 B. NCl_3 and BF_3
C. CO_2 and OF_2 D. NH_3 and PH_3
29. Which of the following molecules does not obey the octet rule? (1)
- A. NCl_3 B. CCl_4
C. CO_2 D. ClO_2

30. Which of the following explains why gases are highly compressible? (1)
- A. Gas particles are tightly packed together.
 - B. Gas particles have weak intermolecular forces.
 - C. Gas particles are far apart and have large empty spaces between them.
 - D. Gas particles lose energy when compressed.
-
31. At what temperature will the volume of a gas be double if its original temperature is 27°C at constant pressure? (1)
- A. 150 K
 - B. 300 K
 - C. 600 K
 - D. 700 K
-
32. The rate of diffusion of methane (CH₄) is twice that of an unknown gas. What is the molecular mass of the unknown gas? (1)
- A. 32
 - B. 64
 - C. 44
 - D. 48
-
33. The intermolecular force of three liquids is given as follow: Liquid A < Liquid B < Liquid C. Which statement is correct? (1)
- A. Molar heat of vaporization: liquid A > Liquid B > liquid C
 - B. Normal boiling point: Liquid B < liquid A < liquid C
 - C. Rate of vaporization: liquid C < Liquid B < liquid A
 - D. Vapor pressure liquid C > Liquid B > liquid A
-
34. The rate of a chemical reaction will be increased by all of the following factors EXCEPT. (1)
- A. the presence of a high activation energy
 - B. increasing the surface area of a reactant
 - C. the presence of a catalyst
 - D. an increase in temperature
-
35. What is the process that converts liquid vegetable oils to solid fats? (1)
- A. Hydration
 - B. Hydrogenation
 - C. Hydrolysis
 - D. Saponification
-
36. The correct relationship between K_c and K_p for the reaction $\text{CO(g)} + \text{H}_2\text{O(g)} \leftrightarrow \text{CO}_2\text{(g)} + \text{H}_2\text{(g)}$ at equilibrium is (1)
- A. $K_p = K_c(RT)^2$
 - B. $K_p = K_c(RT)$
 - C. $K_p = K_c$
 - D. $K_p = K_c(RT)^4$
-
37. When propene, C₃H₆ undergo a complete combustion reaction, what will be the coefficient of oxygen? (1)
- A. 9
 - B. 6
 - C. 4
 - D. 3
-
38. The percentage ionization of a 0.01 mol/L weak acid HA solution whose pH is 4 is (1)
- A. 0.1%
 - B. 1%
 - C. 2.5%
 - D. 4%
-
39. The pH from the titration of weak acid and strong base at the equivalent point is ----- (1)
- A. below 7
 - B. equal to 7
 - C. above 7
 - D. not determined

40. The conjugate base and conjugate acid for hydrogen carbonate ion, HCO_3^- respectively is (1)

A. CO_3^{2-} and HCO_3^- B. H_2CO_3 and CO_3^{2-}
C. CO_3^{2-} and H_2CO_3 D. HCO_3^- and H_2CO_3

41. The dissociation constant of a weak acid is 1×10^{-4} . In order to prepare a buffer solution with a pH = 5, the (acid)/(salt) ratio should be (1)

A. 1:10 B. 4:5
C. 10:1 D. 5:4

42. What is the product for the following reaction given below? $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOCH}_2\text{CH}_3$ with LiAlH_4 (1)

A. $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$ & $\text{CH}_3\text{CH}_2\text{OH}$ B. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ & $\text{CH}_3\text{CH}_2\text{OH}$
C. $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$ & CH_3COOH D. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$ & $\text{CH}_3\text{CH}_2\text{OH}$

43. Which of the following combinations does NOT form a buffer solution? (1)

A. NH_3/HCl in 2:1 mole ratio B. $\text{HNO}_3/\text{KNO}_3$ in 1:1 mole ratio
C. $\text{N}_2\text{H}_4/\text{N}_2\text{H}_5\text{Cl}$ D. $\text{HCOOH}/\text{HCOOK}$

44. Which substance is the reducing agent in the following reaction? $4\text{H}^+(\text{aq}) + 2\text{Cl}^-(\text{aq}) + \text{MnO}_2(\text{s}) \rightarrow \text{Cl}_2(\text{g}) + \text{Mn}^{2+}(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$ (1)

A. $\text{H}^+(\text{aq})$ B. $\text{Cl}^-(\text{aq})$
C. $\text{Cl}_2(\text{g})$ D. $\text{MnO}_2(\text{s})$

45. Which one of the following statements is not true regarding the electrolysis of concentrated sodium chloride solution? (1)

A. Chlorine gas is produced at the anode.
B. Oxygen gas is produced at the anode.
C. Hydrogen gas is formed at the cathode.
D. After the electrolysis, the remaining solution becomes basic.

46. The volume of O_2 formed at STP when Na_2SO_4 is electrolyzed by passing 100A in 965 seconds? (1)

A. 9.6 L B. 22.4 L
C. 5.6 L D. 0.25 L

47. If the reaction proceeds spontaneously, which of the following sets of conditions is satisfied? (1)

A. $\Delta G^\circ < 0$, $K < 1$, $E^\circ < 0$ B. $\Delta G^\circ > 0$, $K > 1$, $E^\circ < 0$
C. $\Delta G^\circ < 0$, $K > 1$, $E^\circ > 0$ D. $\Delta G^\circ > 0$, $K < 1$, $E^\circ < 0$

48. Which one of the following is not true about electroplating of a tray made of iron with chromium? (1)

A. Chromium metal placed in the anode
B. A tray made of iron placed in the cathode
C. Use iron salt solution as an electrolyte
D. Use chromium salt solution as an electrolyte

49. Two cells are connected in series. One contains AgNO_3 and the other contains CuSO_4 . If 8g of Cu is deposited at one electrode, how many grams of Ag is deposited at the other electrode? (molar mass Ag=108 g/mol, Cu=64 g/mol) (1)

A. 4 B. 14
C. 27 D. 54

-
- 50.** Which industrial chemicals mismatched with its manufacturing process? (1)
- | | |
|--|--|
| A. NH ₃ - Haber-Bosch process | B. HNO ₃ - Ostwald process |
| C. H ₂ SO ₄ - Contact process | D. Na ₂ CO ₃ - Frasch process |
-
- 51.** Among the following nitrogen-based fertilizers which is a high percentage nitrogen content? (1)
- | | |
|-----------------------------|---|
| A. Anhydrous ammonia | B. Urea-ammonium nitrate (UAN) solutions |
| C. Urea | D. DAP |
-
- 52.** All of the following undergoes addition polymerization except (1)
- | | |
|------------------------------------|-----------------------|
| A. Polyvinyl chloride (PVC) | B. Teflon |
| C. Nylon | D. Polystyrene |
-
- 53.** The correct order of preparatory stages in the tanning processes of animal hide or skin? (1)
- A.** Soaking → scudding → flesh removal → hair removal → deliming → curing
- B.** Scudding → soaking → flesh removal → hair removal → curing → deliming
- C.** Deliming → flesh removal → hair removal → soaking → scudding → curing
- D.** Curing → soaking → flesh removal → hair removal → scudding → deliming
-
- 54.** What substance is used to bleach the brown syrup in the process of sugar manufacturing? (1)
- | | |
|------------------------------|-------------------------------|
| A. Potassium chlorate | B. Sulfur dioxide |
| C. Calcium chloride | D. Sodium hypochlorite |
-
- 55.** Which one of the following is a thermosetting plastic? (1)
- | | |
|--------------------|------------------------------|
| A. Bakelite | B. Polyethylene |
| C. Dacron | D. Polyvinyl chloride |
-
- 56.** Any discharge of a solid, liquid or gaseous substance or radiation (energy) into an environment that causes unwanted changes is said to be: (1)
- | | |
|------------------------|---------------------|
| A. Pollution | B. Corrosion |
| C. Contaminates | D. Slug |
-
- 57.** The component of the environment consisting earth's upper outermost skin and accessible to human beings is known as _____. (1)
- | | |
|-----------------------|-----------------------|
| A. Atmosphere | B. Hydrosphere |
| C. Lithosphere | D. Biosphere |
-
- 58.** Which of the following is not a greenhouse gas? (1)
- | | |
|---|---|
| A. Carbon dioxide (CO ₂) | B. Nitrogen dioxide (NO ₂) |
| C. Water vapor (H ₂ O) | D. Oxygen (O ₂) |
-
- 59.** Which principle of green chemistry focuses on eliminating waste rather than cleaning it up later? (1)
- | | |
|--|--|
| A. Atom Economy | B. Prevention |
| C. Design for Energy Efficiency | D. Safer Solvents and Auxiliaries |
-
- 60.** Which principle emphasizes maximizing the desired product and minimizing by-products? (1)
- | | |
|---|--------------------------------------|
| A. Design for energy efficiency | B. Use of renewable feedstock |
| C. Less hazardous chemical synthesis | D. Atom economy |

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- 61.** Identify the correct statement about metallic conductivity and electrolytic conductivity. (1)
- A. Metallic conductivity decreases while electrolytic conductivity increases with increasing temperature
 - B. Both conductivities increase with increasing temperature
 - C. Both conductivities decrease with increasing temperature
 - D. Metallic conductivity increases while electrolytic conductivity decreases with increasing temperature
-
- 62.** Equal energy of orbitals (degenerate orbitals) are each occupied by a single electron before the second electron of opposite spin enters the orbital. This principle is (1)
- A. Avogadro's principle
 - B. Aufbau principle
 - C. Pauli's exclusion principle
 - D. Hund's principle
-
- 63.** Which of the following is true about homologous series? The Homologous series is? (1)
- A. The existence of chemical compounds that have the same molecular formula but different in structural formation
 - B. A structure in which the isomers have different functional group with respect to the double bond
 - C. The property of an element to form chains with the repetition of the same atom
 - D. A group of molecules each differing from the next member by a certain group
-
- 64.** Which one of the following is not amphoteric? (1)
- A. HS^-
 - B. NH_4^+
 - C. H_2O
 - D. HCO_3^-
-
- 65.** A solution of H_2SO_4 is made by dissolving 245g of the acid in enough water to produce 500ml solution. What is the normality of the solution? (At. Wt. H=1, S=32, O=16) (1)
- A. 2
 - B. 4
 - C. 8
 - D. 10

— END OF EXAMINATION —